CS 550 Operating Systems, Spring 2026
Homework Assignment 2 (HW2)

Out: 2/19/2026 Thu.
Due: 2/25/2026 Wed. 23:59:59

Full points: 100

Note:

- There are six questions, each on a separate page.
- Read the submission instructions on the last page carefully and prepare the submission accordingly.

1. (24 points) You have a single-core system with timer interrupts every 1 ms. Consider this user program P:

```
int fd_in, fd_out, len;
char * buf[4096];

int main() {
  len = 0;                            // A
  fd_in = open("in.txt", O_RDONLY);   // B
  fd_out = open("in.txt", O_WRONLY);  // C
  read(fd_in, buf, 4096);             // D
  buf[4095] = 0;                      // E
  write(fd_out, buf, 4096);           // F
  return 0;
}
```

Assume the following.

- The `read()` function executes synchronously, meaning it returns when the I/O operation completes, and the desired content is read into the buffer.
- No other I/O activities (other than the ones caused by P) happen during the execution of P.

Determine for each line labeled A to F in the code, whether control can transfer to OS due to the following interrupts or traps during the execution of the line. If so, what interrupt(s) can trigger the transfer?

- Timer interrupt
- I/O completion interrupt
- System call (software) interrupt (i.e., syscall trap)
- Divide overflow (software) interrupt (i.e., divide-by-zero exception)

	Timer Interrupt	I/O completion Interrupt	sys call Interrupt	Divide overflow.
(A)	Yes	No	No	No.
(B)	Yes	Yes	Yes	No
(C)	Yes	Yes	Yes	No
(D)	Yes	Yes	Yes	No
(E)	Yes	No	No	No
(F)	Yes	Yes	Yes	No

2. (25 points) A system has the following settings:

- It uses the STCF (i.e., preemptive SJF) scheduling policy.
- If two jobs with the same TCF (time-to-completion), the one that arrived earlier is schedule first.
- Overhead of switching between two jobs (i.e., context switch overhead) is 0 ms (i.e., no overhead).
- Timer interrupt happens every 1 ms.

A job J with job length 500 ms arrives at time 0. Starting at time 0, there is also infinite stream of short jobs arrive with a 10 ms interval and each job has a job length of 9 ms. **Question**: Will the job J ever finish? If yes, at what time? If not, why?

J can finish.

∵ the short job with length 9ms arrival every 10ms

∴ Every 10ms J can be excuted 1ms.

Also, the sys using STCF, and when 2 jobs have same TCF, the one that arrived earlier is schedule first

∴ when J has 9ms left, it will not be switched to short job.

∴ the total time of finish J is

$491 \times 10 + 9 = 4919$ ms.

3. (26 points) Three jobs, A, B, and C, arrives in a single-CPU system at time 0. They need 20, 30, and 20 seconds of CPU time, respectively. Round-robin (RR) scheduling policy is used in the system. Suppose the scheduler schedules the three jobs in the order of "A-B-C" when they arrive, and there is no cost for context switching. Answer the following questions.

(1) (5 points) At what time does A finish with a one-second long time slice?

(2) (5 points) At what time does B finish with a one-second long time slice?

(3) (16 points) Assuming the time slice length is undetermined, is it possible that B finishes before C? If yes, how can this happen (list all such possibilities)? If no, why?

(1) Every 3 seconds A got 1 second to be called

$\therefore 19 \times 3 + 1 = 57 + 1 = 58\,s$.

(2) Every 3 seconds B got 1 second

$\therefore 29 \times 3 + 2 = 87 + 2 = 89\,s$

(3) Yes, if the time slice is greater or equal than 30s, then B is possibly finished before C, or in the period that can not finish C in one time slide but can finish B in 2 time slide.

$\therefore$ time slide $\in [15, 20) \cup [30, +\infty)$.

4. (25 points) Jobs A, B, C are CPU-bound and their arrival times are unknown. The following is observed:

A: 7ms.
B: 4ms.
C: 1ms.

- A runs first for 2 ms (not done)
- B runs next for 2 ms (not done)
- C runs next and runs to completion for 1 ms
- B runs to completion for 2 ms
- A runs to completion for 5 ms

Question: For each policy below, determine if it can lead to the above runtime behaviors.

(1) FIFO (X).

(2) SJF (X).

(3) STCF (√).

(4) RR (√).

(5) MLFQ (√)

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(1) Not possible, because A running first, but not finished before running B.

(2) not possible, because SJF running the shortest every time, but once job is running it won't be switched until it finished.

(3) possible, because total running time of A is 7ms, B is 4, C is 1, so it is possible in the way like, A comes first, then after 2ms, B come, then A has 5ms left. B has 4ms, so running B. after 2ms C comes, C only has 1ms, B has 2ms left. then do C. after C finished, B has 2ms, A has 5ms. so running B, then A

(4) it is possible, with the time slides 2ms.

(5) it is possible, time slides 2ms,
initial priority is A B C.
then after that priority is B. A.